



Program: BS Computer Science

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Credit Hour(s): 1

Pre-requisite(s): EE1005 Digital Logic Design

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Lab # 14: Delay Loops

1. Objectives

- Understand the concept and necessity of Delay Loops in assembly language.
- Implement Nested Loops to create longer, perceptible delays.
- Manipulate Video Memory to create simple animations controlled by delay subroutines.

2. Delay Loops

Delay loops are used to create fixed time intervals in 8088 assembly. Since the processor executes instructions extremely fast, we use "dummy" loops (instructions that do nothing but waste time) to pause execution. This is critical for:

- Hardware interfacing.
- Display timing (e.g., scrolling text).
- Synchronizing I/O operations.

2.1. Factors Affecting Delay

The actual duration of a delay depends on three main factors:

1. **CPU Clock Frequency:** The speed of the processor (e.g., 4.77 MHz, 5 MHz, etc.)
2. **Instruction Cycle Counts:** How many clock cycles a specific instruction (like DEC or JNZ) takes to execute
3. **Loop Nesting:** The depth of the loop (Single vs. Nested)

2.2. Nested Delay Loop Structure

```
DELAY2:
MOV CX, 200          ; Outer Loop
AGAIN3:
    MOV BX, 200      ; Middle Loop
    AGAIN2:
        MOV DX, 10   ; Inner Loop
        AGAIN1:
            DEC DX
            JNZ AGAIN1
        DEC BX
    
```



```
JNZ AGAIN2  
DEC CX  
JNZ AGAIN3  
RET
```

Example 1: Basic Animation and Single Loop Delay

```
[org 0x0100]  
  
mov ax, 0xb800  
mov es, ax  
  
mov di, 0          ; Start position (Top Left)  
mov cx, 50         ; Move 50 steps  
next_step:  
    mov byte [es:di], 'A'      ; Draw character  
    mov byte [es:di+1], 0x07   ; Draw attribute (White on Black)  
  
    call delay                ; Pause execution  
  
    mov byte [es:di], ' '      ; Clear character (erase)  
    add di, 2                  ; Move to next cell (2 bytes per cell)  
  
    loop next_step  
mov ax, 0x4c00  
int 21h  
  
delay:  
    mov ax, 0ffffh ; Load max value  
d1: dec ax  
    jnz d1          ; Jump if not zero  
    ret
```

Example 2: Implementing Nested Loops for Visible Delay

Replace the delay subroutine in Example 1 with the nested version.

```
delay:  
    mov ax, 200          ; Outer Loop Counter  
outer:  
    mov dx, 65000        ; Inner Loop Counter  
inner:  
    dec dx  
    jnz inner            ; Inner loop logic  
    dec ax  
    jnz outer            ; Outer loop logic  
    ret
```



3. Lab Task

Write an assembly program for the "Bouncing Ball" animation. Apply logic to make the character bounce back when it hits a limit.

Modify your code from Example 2 to achieve the following:

- 1) Move the character 'O' from left to right for 20 steps.
- 2) Once it reaches step 20, move it **backwards** (right to left) to the starting position.
- 3) **Hint:** You will need two loops in your main program.
 - Loop 1: add di, 2 (Move Right)
 - Loop 2: sub di, 2 (Move Left)